

Group By:

```
mysql> SELECT department, COUNT(*) AS total_students  
-> FROM student  
-> GROUP BY department;
```

department	total_students
CSE	2
ECE	1
IT	1

3 rows in set (0.02 sec)

```
mysql> SELECT department, AVG(marks) AS average_marks  
-> FROM student  
-> GROUP BY department;
```

department	average_marks
CSE	86.5000
ECE	78.0000
IT	92.0000

3 rows in set (0.01 sec)

Having:

```
mysql> SELECT department, AVG(marks) AS average_marks  
-> FROM student  
-> GROUP BY department  
-> HAVING AVG(marks) > 80;
```

department	average_marks
CSE	86.5000
IT	92.0000

2 rows in set (0.01 sec)

Order By:

```
mysql> SELECT *
-> FROM student
-> ORDER BY marks ASC;
```

student_id	name	department	marks
102	Bala	ECE	78
101	Arun	CSE	85
104	Ravi	CSE	88
103	Kavin	IT	92

4 rows in set (0.01 sec)

```
mysql> SELECT *
-> FROM student
-> ORDER BY marks DESC;
```

student_id	name	department	marks
103	Kavin	IT	92
104	Ravi	CSE	88
101	Arun	CSE	85
102	Bala	ECE	78

4 rows in set (0.00 sec)

All combined:

```
mysql> SELECT department, AVG(marks) AS average_marks
-> FROM student
-> GROUP BY department
-> HAVING AVG(marks) > 80
-> ORDER BY average_marks DESC;
```

department	average_marks
IT	92.0000
CSE	86.5000

2 rows in set (0.00 sec)